EXECUTIVE PROFILE · APPLIED AI

Research depth. Product judgement. Strategic reach.

Bruno Trentini is an Applied ML Research Scientist at NVIDIA and Oxford PhD candidate whose career spans AI research, technical product delivery, GPU platforms, research alliances, startup ecosystems and customer-facing programmes.

EXECUTIVE SUMMARY

Applied AI leadership at the interfaces

Bruno works where research, engineering, product, customers, partners and executives meet. His differentiator is the ability to remain technically credible while aligning specialists, translating emerging methods into adoption paths and representing complex AI programmes to external scientific and industry audiences.

His recent work combines generative and geometric machine learning with biomolecular AI, GPU/HPC workflows and research-to-product programmes. Earlier roles provide a grounding in production ML, product management, commercial analytics, consulting and software engineering.

BEST-FIT MANDATE

Scope rather than title

Applied AI and strategy

Technical direction, portfolio choices and translation of emerging AI into programmes and platforms.

Research and engineering

Leadership of researchers, ML engineers and technical specialists around measurable delivery.

Strategic engagements

Executive, customer and partner interaction that remains grounded in technical substance.

Research ecosystems

Alliances, academic partnerships, workshops and external representation across AI for Science.

The strongest fit is a Director, Head or well-designed VP mandate spanning Applied AI, AI strategy and innovation, AI solutions engineering, applied research, AI products, research alliances, strategic AI partnerships or AI for Science.

SELECTED EXPERIENCE

Leadership and delivery

SEP 2024–PRESENT

Applied ML Research Scientist

NVIDIA

- Design generative-model architectures and efficient samplers, including entropy-rate scheduling with an 18.1% MMD improvement and a paired 22.7% SDE-Heun gain across benchmark settings.
- First author of Neural Point-Forms, connecting diffusion geometry and discrete exterior calculus with high-dimensional biological point-cloud analysis.
- Partner with product and engineering teams on BioNeMo Framework and NIMs, contributing evidence, examples and multi-node workflows that help users move from model access to reproducible GPU use.
- Continue to lead and contribute to scientific workshops across ICML, ICLR, NeurIPS and EurIPS.

JAN 2023–AUG 2024

Global Life Sciences Research Alliances Lead

NVIDIA

- Led an eight-person science and engineering effort on protein inverse folding for peptide design, improving sequence diversity by 20% while maintaining structural similarity.
- Coordinated applied AI programmes with life-sciences partners across biomolecular models, GPU/HPC workflows and multi-node training; supported examples reporting up to 11.3× task speed-up and 49% lower cost.
- Oversaw startup and academic projects on Cambridge-1 readiness, coordinating technical engagement and model optimisation with partner teams.
- Built AI-for-science workshop and community programmes connecting senior researchers, founders and applied ML teams.

MAY–DEC 2022

Developer Relations Manager — Healthcare & Life Sciences Startups Lead, EMEA

NVIDIA

- Led regional engagement with healthcare and life-sciences startups, helping technical teams scope GPU-accelerated ML workflows, demonstrations and partner examples.
- Helped run healthcare developer summit programming reaching 2,000+ developers.
- Worked with founders, technical leaders and product stakeholders on adoption, partnerships and technical roadmap decisions.

OCT 2020–MAY 2022

Senior Data Scientist; Product Manager — AI & Analytics

GSK

- Built and deployed a reinforcement-learning and Bayesian campaign-allocation system that reduced cPCV by 55% in A/B testing and was shortlisted in the top three at the DataIQ awards.
- Owned full-stack implementation across Spark/Databricks, PyTorch, APIs, Azure and an operational HTML/JavaScript dashboard.
- Mentored five apprentices, directly managed two and educated 100+ colleagues through technical seminars.
- Previously led three data engineers, established OKRs and supported regional analytics rollout.

DEC 2018–DEC 2019

AI Startups Ecosystem Manager

NVIDIA

- Advised 100+ AI startups on GPU-accelerated ML workflows through technical sessions, workshops and ecosystem programmes.
- Built relationships with accelerators and startup programmes across Europe and ran founder and VC events covering applied AI, HPC and developer enablement.

2008–2020

Earlier product, analytics, consulting and software career

Hotelbeds · Alliants · GVT/Telefónica · KIT-AR/UCL · EY · HSBC

- Built commercial analytics tools used 1,000+ times, forecasting systems and an early ML-powered churn capability for global account teams.
- Delivered predictive systems and product-launch support in client consulting, including the first Four Seasons mobile application.
- Led innovation-lab computer-vision work, contributed to technical roadmaps and engaged executive leadership on strategic initiatives.
- Began in global banking software and IT consulting, building a durable grounding in enterprise systems and operational delivery.

SELECTED EVIDENCE

Research, platforms and external leadership

First-author research

Neural Point-Forms and Entropy Across the Bridge, spanning geometric learning and efficient generative sampling.

Research-to-product

BioNeMo Framework and NIM work connecting scientific models with reproducible GPU and multi-node workflows.

Scientific programmes

Organiser and contributor across ICML, ICLR, NeurIPS and EurIPS workshops in biology, chemistry and materials.

External representation

Technical talks, industry forums, academic seminars, developer summits and founder-facing events.

EDUCATION

Scientific foundation

PhD in Computer Science, Machine Learning

University of Oxford · 2024—2028
Generative Models, Geometric Deep Learning and AI for Science.
Supervisor: Prof. Michael Bronstein.

MSc Data Science & Machine Learning

University College London · Distinction
Deep Learning, Reinforcement Learning, Probabilistic Learning and Graphical Models.

Postgraduate Diploma in Telematics

Federal University of Technology — Paraná
Distributed systems, networking and programming; monograph awarded 10/10.

BSc Information Systems

Pontifical Catholic University of Paraná
BCI/CREPUQ scholarship and international study in Canada and France.

CONTACT

Bruno Trentini

Based in the United Kingdom. Available for conversations about applied AI leadership, AI for Science, research and engineering programmes, strategic engagements and research ecosystems.

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